



Maulana Azad National Institute of Technology, Bhopal
Department of Electrical Engineering
End Term Examination, May 2025
B.Tech. IV Semester

Subject: Electrical Machines-II

Max. Marks: 50

Subject Code: EE-222

Time: 3 Hours

Note: All Questions are compulsory. Attempt the questions in sequence.

Q.No.	Question	Marks	CO
1. a.	A 3-phase, 50 Hz, 8-pole induction motor has a slip of 6%. Find (i) Speed of the motor (ii) Frequency of the rotor emf. If the rotor has a resistance of 2 ohms and standstill reactance of 4 ohms, find the power factor at (i) standstill position at speed of 650 rpm.	05	CO1
1. b.	What are the methods used to reduce the starting current of a three-phase induction motor? Explain clearly with the help of equations.	05	CO2
2. a.	A 500 V, 3-phase, star connected induction motor has the following impedances per phase referred to the stator winding: Stator:- $(0.5+j1)$ ohms and rotor:- $(0.7+j1)$ ohms and magnetizing branch:- $(10+j50)$ ohms. Find (i) Maximum torque (ii) Slip at which maximum torque is produced.	05	CO5
2.b.	Why is a three-phase induction motor self-starting whereas a single-phase induction motor is not? Support your answer with suitable phasors and waveforms.	05	CO4
3. a.	Explain the construction and working of a capacitor-start single-phase induction motor. What are its advantages over a split-phase motor?	05	CO4
3.b.	A 3-phase, 415 V, star connected, induction motor gives the following test results during the test: No load test: 415 V, 20 A, $W_1=5200$ W, $W_2=-3200$ W Blocked rotor test: 60 V, 60 A, $W_1=2200$ W, $W_2=700$ W Find the circuit parameters.	05	CO2
4.a.	A 3-phase, 11 kV, 100 MVA synchronous generator has a synchronous reactance of 1.5Ω per phase. Calculate the terminal voltage when delivering full load at (i) 0.8 power factor lagging and (ii) unity power factor.	05	CO3
4.b.	What is armature reaction in synchronous machines? How does it vary with different load power factors?	05	CO3
5.a.	The following are the parameters of a 230V, 50 Hz, 4-pole, single phase induction motor $R_1=2.2\Omega$; $X_1=3.0\Omega$; $R_2=3.8\Omega$; $X_2=2.1\Omega$; $X_m=86\Omega$. Calculate the input current and power when the motor is operating at full-load speed of 1410 rpm. If the mechanical losses (iron and friction loss) are 60 W, determine gross power developed, useful power at the shaft, shaft torque and efficiency.	05	CO5
5.b.	What are V-curves and inverted V-curves of a synchronous motor? What information do they convey?	05	CO3